

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457698.6341 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1411 +/- 0.0096
Transit depth (Rp/Rs)^2	1.99 +/- 0.27 [%]
Orbital Inclination (inc)	88.61 +/- 0.86
Airmass coefficient 1 (a1)	1.244 +/- 0.013
Airmass coefficient 2 (a2)	-0.1051 +/- 0.0044
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	None
Optimal Aperture	3.62
Optimal Annulus	5.02
Transit Duration (day)	0.1076 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1411 $\pm$ 0.0096 vs 0.1446 (deviation 0.36 σ)
Duration fit vs published	0.1076 d vs 0.1087 d (deviation 0.24 σ)
Mid-time O-C	-5.8 min
Depth SNR	14.7
Residual scatter	1.05 %

Light curve

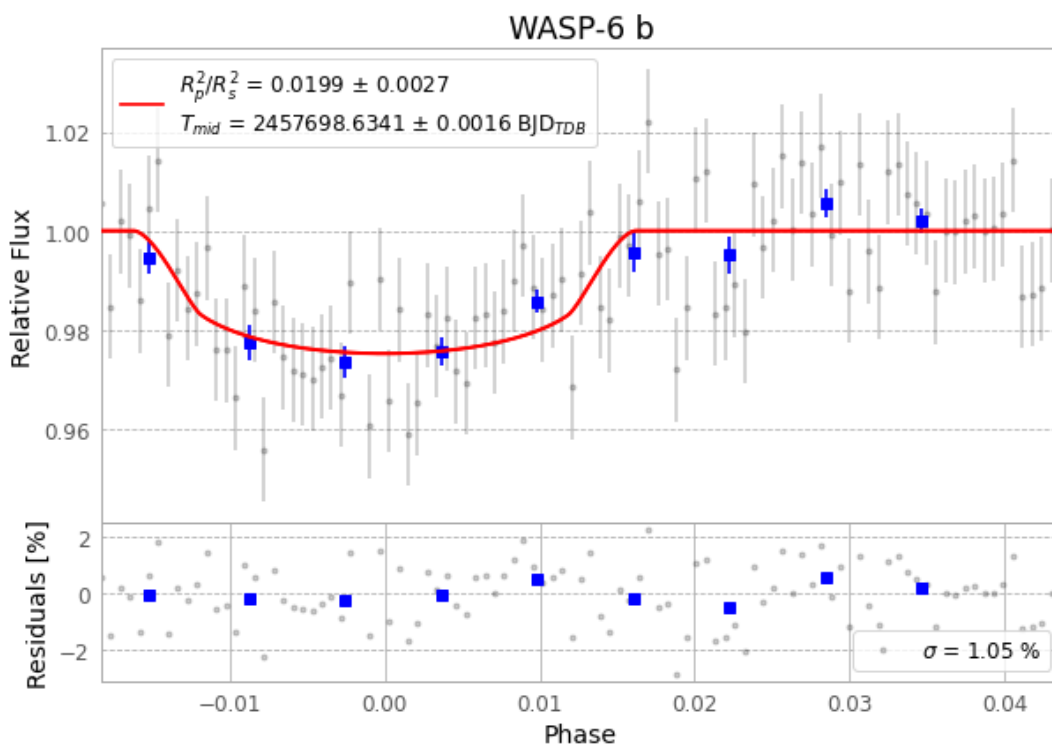


Figure 1 — FinalLightCurve_WASP-6 b_2016-11-05.png (SHA-256 ce9667153e431225...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [272, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a08a39c398bfec78...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

101 FITS frames (66 MB), WASP-6161106013912.FITS → WASP-6161106063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-6-161106.log (def26d47aad8...), FinalLightCurve_WASP-6 b_2016-11-05.png (ce9667153e43...), FinalLightCurve_WASP-6 b_2016-11-05.csv (c6485a95ee2f...)

Compute resources

Wall time 182.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:08Z.